

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- Small group office hours continue through this week
- Last requizzing session this Friday, 3-5pm in **Eckhart 206**.
- Flex (or flex plan) due by end-of-day Friday
- Final Exam review sessions: Saturday, 3:00-4:30pm, **Kent 120** (packet)
Sunday, 3:00-4:30pm, **Kent 120** (2021 final exam)
- End-of-quarter showcase: Monday, 3:00-4:30pm, **Eckhart 202**
Sign up to present by end-of-day on Friday.
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am

Typos in review packet:

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a unitary matrix and T is an upper triangular matrix. In the case of real square matrices, any real square matrix A with all real eigenvalues can be represented as $A = UTU^{-1}$, where U is a real orthogonal matrix and T is a real upper triangular matrix.

Cayley–Hamilton Theorem (Thm. 9.1.1). Let A be a square matrix, and let $p(\lambda) = \det(A - \lambda I)$ be its characteristic polynomial. Then $p(A) = 0$, where 0 is the zero matrix.

$$\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}^2 - 5 \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} + 6 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 10 \\ -5 & 14 \end{pmatrix} - \begin{pmatrix} 5 & 10 \\ -5 & 20 \end{pmatrix} + \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{vmatrix} = (1-\lambda)(4-\lambda) + 2 = \lambda^2 - 5\lambda + 6$$

Is every square matrix a “zero” of its characteristic polynomial?

$$\begin{aligned}
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&= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \left| \begin{array}{cc} 1-\lambda & 2 \\ -1 & 4-\lambda \end{array} \right| = (1-\lambda)(4-\lambda) + 2 = \lambda^2 - 5\lambda + 6
\end{aligned}$$

Is every square matrix a “zero” of its characteristic polynomial?

Is every square matrix a “zero” of some (non-constant) polynomial?

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Is every square matrix a “zero” of its characteristic polynomial?

Is every square matrix a “zero” of some (non-constant) polynomial?

For a given square matrix, for what polynomials is it a “zero”?

We have that $M_{n \times n}^{\mathbb{F}}$ is a vector space of dimension n^2 .

Is every square matrix a “zero” of some (non-constant) polynomial?

We have that $M_{n \times n}^{\mathbb{F}}$ is a vector space of dimension n^2 .

This implies that every subset of $M_{n \times n}^{\mathbb{F}}$ of size $n^2 + 1$ is linearly dependent.

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For $A \in M_{n \times n}^{\mathbb{F}}$, consider the set $\{I, A, A^2, \dots, A^{n^2}\}$; it is linearly dependent.

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Therefore there exists scalars $k_0, \dots, k_{n^2}$ such that

$$k_{n^2}A^{n^2} + \dots + k_1A + k_0I = Z$$

where Z is the $n \times n$ zero matrix. ■

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Cayley–Hamilton Theorem:
Every square matrix is a “zero”
of its characteristic polynomial.

(degree n)

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diagonal matrices?

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diagonal matrices?

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$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

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diagonalizable matrices?

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diagonal matrices



diagonalizable matrices?

$$\begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{-1}$$

diagonal matrices



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$$(1 - \lambda)(4 - \lambda) + 2 = \lambda^2 - 5\lambda + 4 + 2$$

$$= \lambda^2 - 5\lambda + 6$$

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$$\begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a + 2b \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \checkmark$$

diagonal matrices 

diagonalizable matrices 

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 2 & 3 & 4 \\ 0 & 4 & 6 & 7 \\ 0 & 0 & 7 & 9 \\ 0 & 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} -4 & 2 & 3 & 4 \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 3 & 9 \\ 0 & 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} -7 & 2 & 3 & 4 \\ 0 & -3 & 6 & 7 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -9 & 2 & 3 & 4 \\ 0 & -5 & 6 & 7 \\ 0 & 0 & -2 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} =$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 2 & 3 & 4 \\ 0 & 4 & 6 & 7 \\ 0 & 0 & 7 & 9 \\ 0 & 0 & 0 & 9 \end{pmatrix} \begin{pmatrix} -4 & 2 & 3 & 4 \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 3 & 9 \\ 0 & 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} -7 & 2 & 3 & 4 \\ 0 & -3 & 6 & 7 \\ 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -9 & 2 & 3 & 4 \\ 0 & -5 & 6 & 7 \\ 0 & 0 & -2 & 9 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} =$$

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upper triangular matrices?

$$\begin{pmatrix} 5 & 2 \\ 0 & 4 \end{pmatrix} \quad (5 - \lambda)(4 - \lambda) = \lambda^2 - 9\lambda + 20$$

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diagonalizable matrices 

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But by the upper triangular (Schur) representation,
all matrices are similar to an upper triangular matrix. ■

diagonal matrices



diagonalizable matrices



upper triangular matrices?



upper triangular matrices?



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$$(5 - \lambda)(4 - \lambda) = \lambda^2 - 9\lambda + 20$$

$$(A - 5I)(A - 4I)$$

Cayley–Hamilton Theorem:
Every square matrix is a “zero”
of its characteristic polynomial.

$$\begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a + 2b \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



But by the upper triangular (Schur) representation,
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diagonal matrices



diagonalizable matrices



upper triangular matrices?



$$\begin{pmatrix} 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

strictly upper triangular $\Rightarrow$ nilpotent

(has a power
that's the
zero matrix)

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Pf. characteristic polynomial is $\pm \lambda^n$

Cayley–Hamilton $A^n = 0$. ■

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nilpotent $\Rightarrow$ similar to a strictly upper triangular matrix

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nilpotent $\Rightarrow$ similar to a strictly upper triangular matrix

Note: nilpotent $\nRightarrow$ strictly upper triangular

Next up:

- Small group office hours continue through this week
- Last requizzing session this Friday, 3-5pm in **Eckhart 206**.
- Flex (or flex plan) due by end-of-day Friday
- Final Exam review sessions: Saturday, 3:00-4:30pm, **Kent 120** (packet)
Sunday, 3:00-4:30pm, **Kent 120** (2021 final exam)
- End-of-quarter showcase: Monday, 3:00-4:30pm, **Eckhart 202**
Sign up to present by end-of-day on Friday.
- Final Exam: Both exams will be held in our classroom, Ryerson 276.
Section 45 (11:30am section) - Thursday, May 25, 5:30-7:30pm
Section 55 (12:30pm section) - Wednesday, May 24, 7:30-9:30am